

Telecommunication and Network Technology: A Student Perspective

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Preface

This book was written based on author's personal lecture notes on the course CND3200 (Computer Network and Communication) in Universiti Putra Malaysia. The purpose of this book is to help first-year Computer Science students, especially those in their second semester to grasp the fundamental concepts and build an intuitive understanding of the CND3200 course. To be honest, the author found the course quite challenging when taking it back in 2025. Unlike other first-year courses such as 'Discrete Structure' or 'Problem Solving and Programming Concepts', this networking course involves a mix of memorization and calculation, making it difficult to decide whether to focus on the memorization aspects or the calculations. Readers do not need advanced mathematical knowledge to follow this book; rather, strong logic and consistent practice are the keys to develop a solid intuition for the material. The author hopes that this book will help readers understand exactly what are they studying and feel confident when walking into the final exam hall. The author would like to express sincere gratitude Mohd. Noor Derahman for his guidance in teaching the course CND3200, which served as an important source of inspiration for this work.

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München, 13 June 2026

Chapter 1

Intro. to telecommunication

Telecommunication is communication over a distance. To understand it better, telecommunication shares similarities with face-to-face conversation: there is a sender, a receiver, a medium, a protocol, and information that is sent and received. The key difference is that telecommunication allows us to exchange information without meeting in person, even when the parties involved are far apart. However later in this book, the focus is on digital telecommunication. Especially the use of technologies like computers and smartphones to communicate over distances. Furthermore, telecommunication involves rules that must be followed to ensure messages are transmitted accurately and correctly (for instance, ensuring the message remains unchanged and does not reach the wrong recipient). Thus, we can identify five essential components of telecommunication:

- a) Message: The data itself.
- b) Sender: The person or device sending the message.
- c) Receiver: The person or device receiving the message.
- d) Medium: For example, air for face-to-face conversation or wired-medium for computer-to-computer communication.
- e) Protocol: A set of rules that must be followed to ensure smooth communication. An analogy is the use of grammar in face-to-face communication.

What is meant by 'data transmitted via telecommunication'?

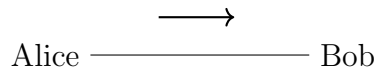
The answer is any type of information that can be sent or received by a computer, such as:

- a) Text
- b) Numbers
- c) Images
- d) Audio
- e) Video

1.1 Transmission Mode

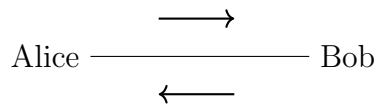
In a physical communication, it is impossible for two people to converse simultaneously, right? The same principle applies to telecommunication. Transmission refers to the rules defining who may send messages, who may receive them, and the specific times for sending or receiving can occur. Telecommunication involves three transmission modes:

- a) Simplex: This is the simplest transmission mechanism, involving exactly only a sender and a receiver.



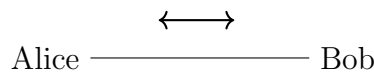
Alice can only send and Bob can only receive, Alice cannot receive and Bob cannot send.

- b) Duplex: This mechanism allows two nodes to both send and receive.



Alice can send to Bob and vice versa, but cannot send at the same time.

- c) Full Duplex: Allows the sender and receiver to communicate simultaneously.



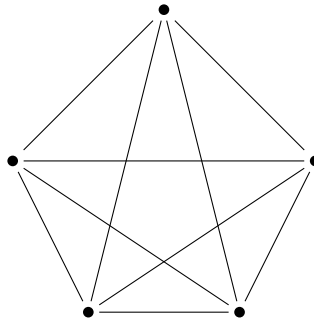
Alice and Bob can send and receive at the same time.

1.2 Network Topology

In this context, “network” refers to a suitable designated telecommunication environment as the number of telecommunication participants grows. There are various topology models, each with its own advantages and disadvantages.

- a) Mesh: In this topology, every node in the network has a direct connection (edge) to every other node without exception. Because this architecture includes connections for every possible path, the cost of implementing it is higher than that of other architectures.

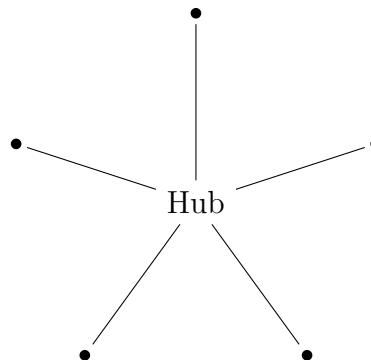
The number of edge for n Nodes is $\frac{n(n-1)}{2}$



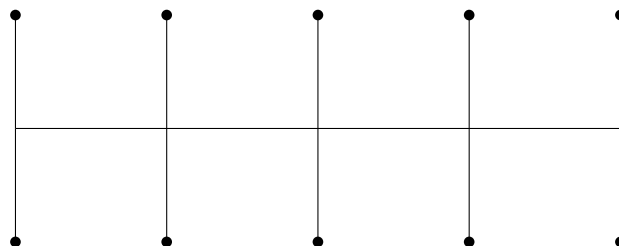
For instance, if $n = 5$, then

$$\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$$

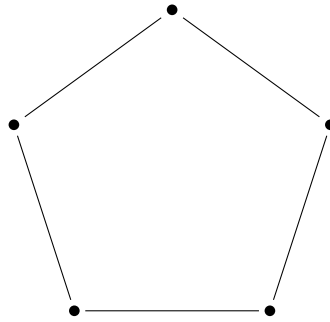
- b) Star: All nodes in the network are connected to a single central hub that acts as an information relay. The advantage of this architecture is that if 1 edge fails, it will not affect the connections between other edges.



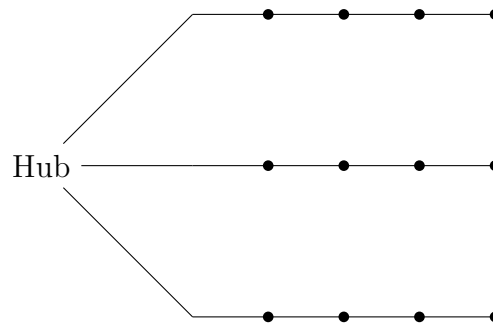
- c) Bus: Each node is connected by a “major road” that links to a specific node. If a section of the major road is damaged, the connection between the nodes is severed.



- d) Ring: A simple connection that forms a loop



- e) Hybrid: The combination of two or more architectures that results in a new, integrated architecture. The following is an illustration of a combined architecture featuring star and bus topologies.



1.3 Network Categories

After the network, what's next? When the number of devices are getting larger, and those devices are connected to 1 another to form a network, it turns out that networks fall into three categories, namely:

- a) Local Area Network (LAN): A network with the smallest coverage area, typically set up by a private single household. Example: If a household has 1 laptop, 1 tablet, and 1 smartphone, and all three devices are connected to each other, this is an example of a Local Area Network with a specific topology.
- b) Metropolitan Area Network (MAN): This is also a type of network similar to a LAN, but with a wider coverage area (for example, covering an entire city).
- c) Wide Area Network (WAN): The largest category of networks. All electronic device in the world that supports telecommunication are part of a WAN.

Now that we know the definition of a network—from topology to categories—two or more connected networks are called the Internet (internetwork).

1.4 Internet Protocol

A protocol is a set of rules that must be followed so that telecommunication can run smoothly. Let's consider an analogy, when we communicate in person (face-to-face), we are actually following a protocol for speaking; such as choosing a language that our conversation partner can understand. The same applies to telecommunication. For instance, if we send a video to another computer but there is no standard protocol, the video will simply appear as random bits to the receiving computer. Leading global organizations such as the IEEE have made significant contributions by developing protocols that set standards for digital communication worldwide.

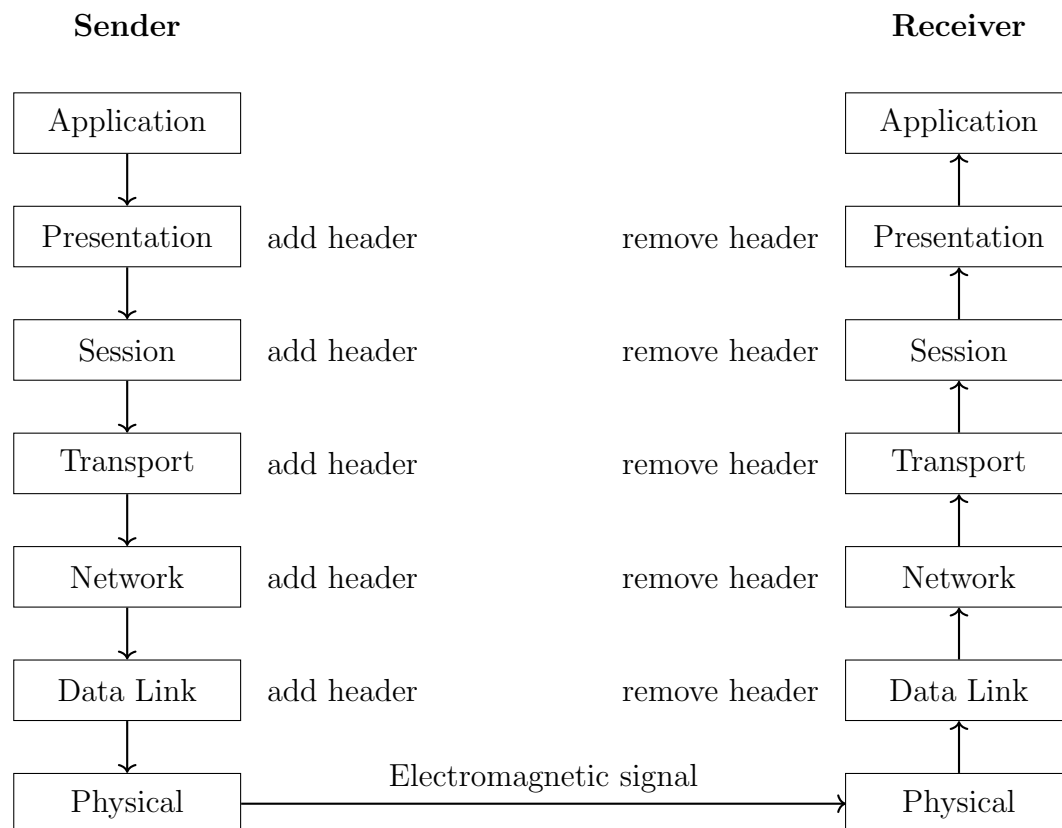
Chapter 2

OSI Model

At the end of Chapter 1, we briefly discussed the protocols that serve as the rules enabling devices to communicate. The OSI model is an example of a protocol with seven layers: Application, Presentation, Session, Transport, Network, Data Link, and Physical. It's easy to remember, just take the first letter of (All People Seem To Need Data Physical). However, it's important to note that:

$$\textit{Application} \longrightarrow \textit{Physical} \neq \textit{Physical} \longrightarrow \textit{Application}$$

Because the same layer can have different functions depending on whether it is sending or receiving. Here is an illustration of how data flows in telecommunication:



The following are the 7 layers of the OSI model, with explanations of their functions relative to the sender:

- i)** Application: Media or software for accessing the internet (such as Instagram and TikTok).
- ii)** Presentation: Manipulating the data to be sent so that the message cannot be viewed by third parties (encryption) and performing compression.
- iii)** Session: Ensuring that the session between the sender and receiver is maintained.
- iv)** Transport: Port configuration and process-to-process communication (see chapter 11).
- v)** Network: To configure the shortest logical path to make data transmission more efficient.
- vi)** Data Link: Access control, flow control, and error control (see Chapter 6).
- vii)** Physical: Converting discrete data into electromagnetic waves and transmitting it continuously.

7 layer of the OSI model with explanations of their functions relative to the receiver:

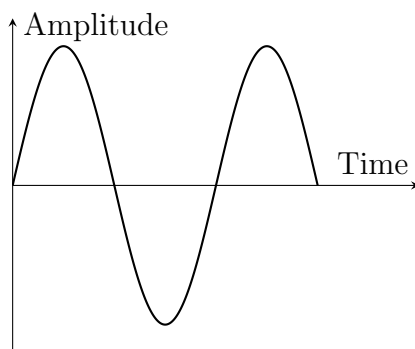
- i)** Physical: Data is received from other devices as electromagnetic waves, and this layer converts them into bits (0s and 1s).
- ii)** Data Link: Organizes the bits into frames and performs error checking (see Chapter 6).
- iii)** Network: Evaluates the path the data has traveled and reads the IP address.
- iv)** Transport: Ensures that process-to-process communication runs smoothly (see Chapter 11).
- v)** Session: Ensures that the session remains connected.
- vi)** Presentation: Decrypts the received data.
- vii)** Application: Displays the received data.

Chapter 3

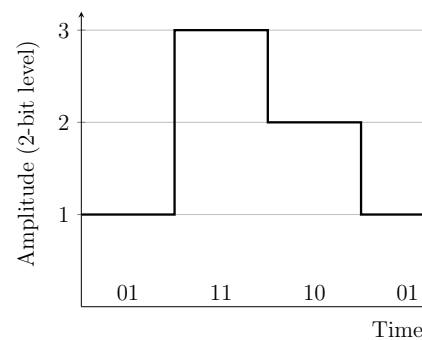
Signal Processing

This chapter may be quite intimidating for some readers because it is dominated by calculations. But readers shouldn't worry regarding the mathematical prerequisites such as Fourier transform or Integral calculus. However, prior knowledge is beneficial.

Firstly, let's define what a signal is. A signal is a concept in graph (usually in $\mathbb{R}^2$) where the amplitude (magnitude) constantly increasing and decreasing over time. Signals are crucial in telecommunication studies because to send a data, the device should prepare the data and converts it into a signal and transmitted via wired or wireless medium (via electromagnetic waves). Signals are divided into two types: analog (continuous) and digital (discrete).



Analog signal example
 $x(t) = \sin t$



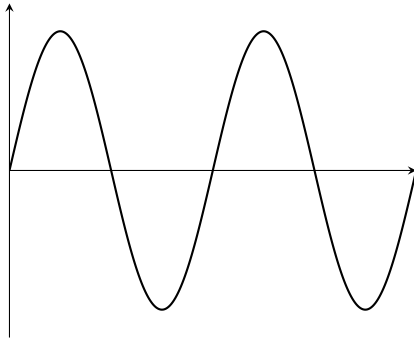
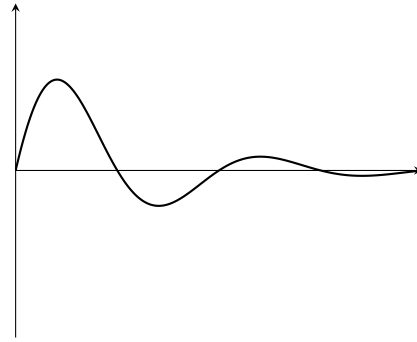
Digital signal example
 $x[t] \in \{00, 01, 10, 11\}$ (2-bit, 4 level)

The main difference between analog signal and digital signal is that, analog signals are more susceptible to interference than digital signals.

Characteristics of Signal

Every signal have 2 characteristics, that is:

- a) Periodic: A signal with a repeating cycle.
- b) Aperiodic: A signal without a repeating cycle.

Periodic Signal $x(t) = \sin t$ Aperiodic Signal $x(t) = e^{-0.3t} \sin t$

Note that periodic signals forms the same pattern even if we extend the x-axis to $\pm\infty$, whereas aperiodic signals do not exhibit the same pattern and, in this case, can be denoted as

$$\lim_{t \rightarrow \infty} e^{-0.3t} \sin t = 0$$

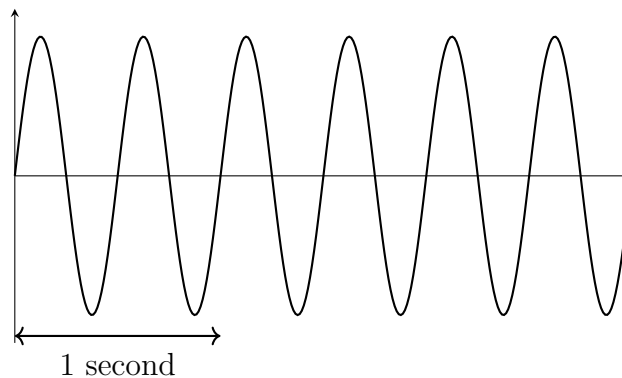
3.1 Period and Frequency

Period and frequency are very basic yet very essential properties of a signal. Period and frequency are inverses of each other and are denoted by:

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

Where f is the frequency and T is time. Frequency refers to the number of a signal completes 1 full cycle in a unit of time (usually seconds or milliseconds). Time on the other hand, refers to the amount of time taken to complete 1 cycle.

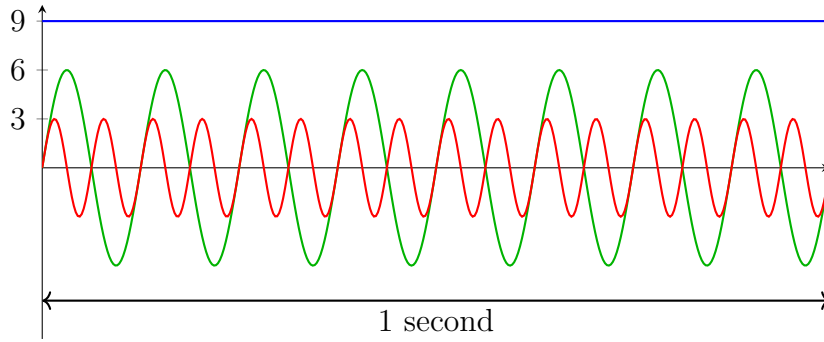
Now let's consider the graph:



If we know that 1 second corresponds to 2 cycles (2 up-and-down movements), then the frequency is $2Hz$ because the cycle occurs twice in 1 second. To find the time T , we substitute this into the following equation:

$$T = \frac{1}{f} \quad \rightarrow \quad T = \frac{1}{2} \quad \rightarrow \quad T = 0.5$$

What this means is that it takes 0.5 seconds for the signal to complete 1 cycle. The following graph represents three signals with different frequencies and amplitudes over a time interval of $T = 1$ second.



If we plot those three signals above with the y-axis representing frequency (in Hertz), then we will obtain a Frequency-Domain Graph which equivalent to the graph shown below.



Composite Signal

A composite signal is a new signal obtained by accumulating or summing all existing signals. The formal formula for a composite signal is:

$$c(t) = \sum_{i=1}^n x_i(t)$$

If n is the number of signals, and $x_i(t)$ is the signal function, denoted by $x_i(t) = \alpha \sin(\beta t)$.

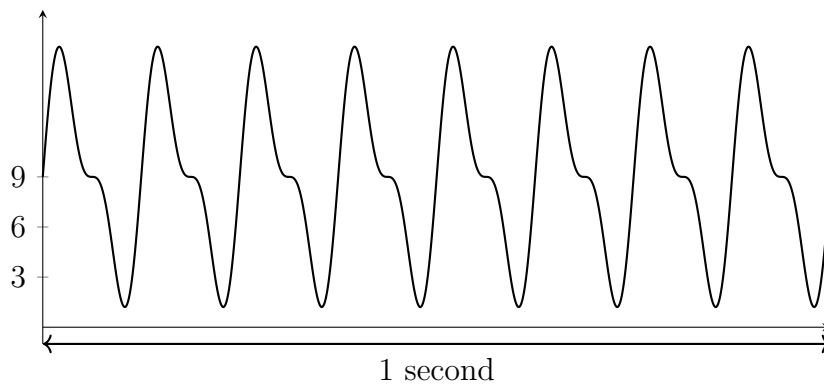
To calculate the composite signal of the 3 signals we had earlier using, let

- i) Blue function: $x_1(t) = 9$
- ii) Green Function: $x_2(t) = 6 \sin 8t$
- iii) Red function: $x_3(t) = 3 \sin(16t)$

Hence:

$$\sum_{i=1}^3 x_i(t) = 9 + 6 \sin 8t + 3 \sin(16t)$$

If we plot the resulted function of $x(t) = 9 + 6 \sin 8t + 3 \sin(16t)$, then we will obtain a new following composite signal:



Note that the composition of periodic signals also produces a periodic signal.

Now that we had understand composite signals, we can define bandwidth. Bandwidth is the difference between the highest frequency and the lowest frequency in a Frequency-Domain Graph (in Hz). Let's take the example of the graphs $x_1(t)$, $x_2(t)$, and $x_3(t)$.

Frekuensi of $x_1(t) : 0Hz$
 Frekuensi of $x_2(t) : 8Hz$
 Frekuensi of $x_3(t) : 16Hz$

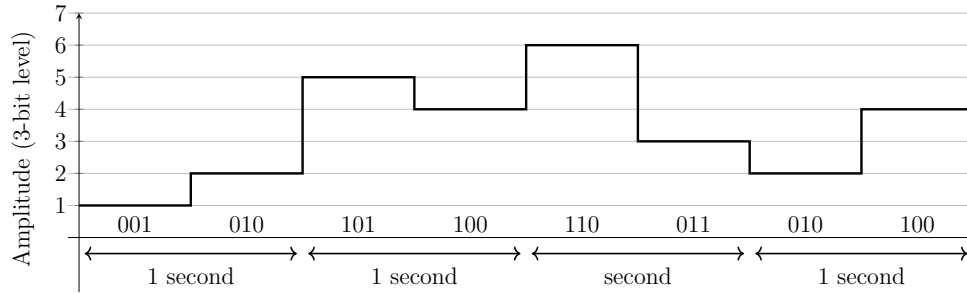
$$16Hz - 0Hz = 16Hz$$

Hence, the bandwidth of the composite signal $x_1(t)$, $x_2(t)$ and $x_3(t)$ is $16Hz$.

Note that non-composite signals are sometimes useless in telecommunication, so we need a combination of more than 1 wavefunction with varying amplitudes and frequencies.

3.2 Digital Signal

As we discussed earlier, a digital signal is a discrete signal. From an electrical engineering perspective, a '1' valued data is represented as a positive voltage, and '0' valued data is represented as a negative voltage (in binary form). This means that the information generated over a given time interval is limited. Consider the graph below as an example:



The graph above shows that every 1 second, the signal carries 6 bits of data.

If there are 8 levels, then each signal is represented by 3 bits, this is because $\log_2 8 = 3$

Transmission Impairment

Information transmitted through a medium such as a cable will absolutely experience energy-loss. This means that the signal at the end of the medium will be different from the signal at the beginning. This energy loss is caused by the resistance of the medium, such as a cable; and is known as attenuation (see electrical physics for more details). The energy loss in the signal can be calculated using the following formula:

$$dB = 10 \log_{10} \left(\frac{P_1}{P_2} \right)$$

Where

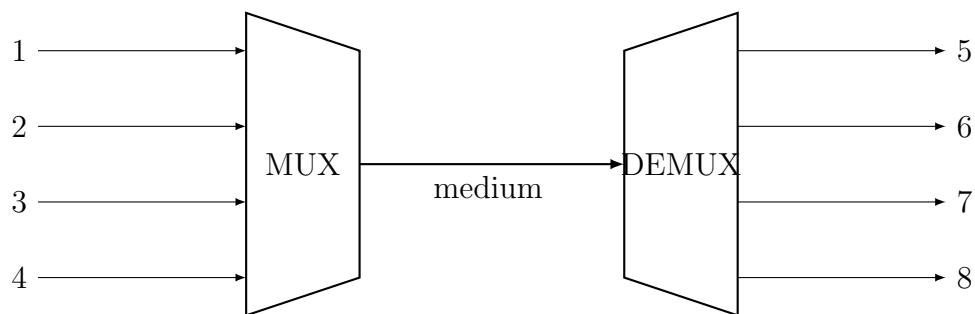
- i) dB is the total energy-loss.
- ii) P_1 is the amount of energy at the beginning of the medium (in Joules)
- iii) P_2 is the amount of energy at the end of the medium (in Joules)

Note: If you look back at the illustration of the OSI model in Chapter 2, the physical layer at the sender converts the data from bits into electromagnetic signals before it is actually transmitted to the receiver. These electromagnetic signals are the analog waves we discussed earlier in Chapter 3.

Chapter 4

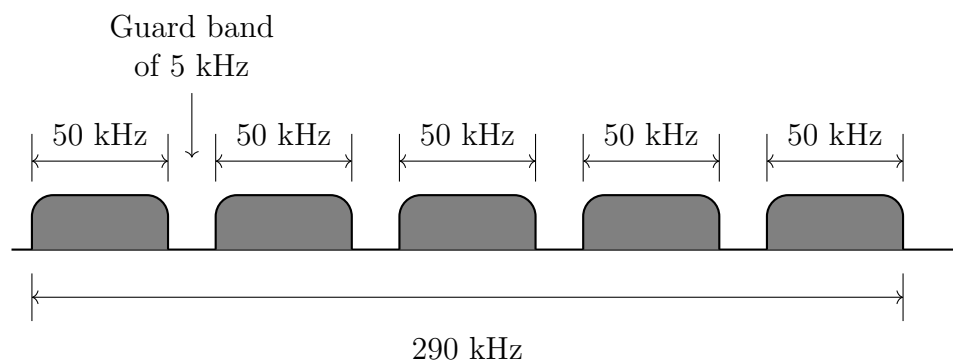
Multiplexing

Multiplexing is a technique that allows multiple devices to transmit data by sharing a single medium. Below is an illustration representing 8 nodes with shared medium, where node number 1-4 are senders and 5-8 are receivers:



4.1 Frequency-Division Multiplexing

Now that we want to manage telecommunication traffic over the shared medium. The first method for doing this is to divide the medium's bandwidth into small segments. As we saw in the previous chapter, bandwidth is an interval on a frequency-domain graph defined by the difference in frequency between the highest and lowest waves. The following illustration shows an example of bandwidth in the cable we are considering as the medium:



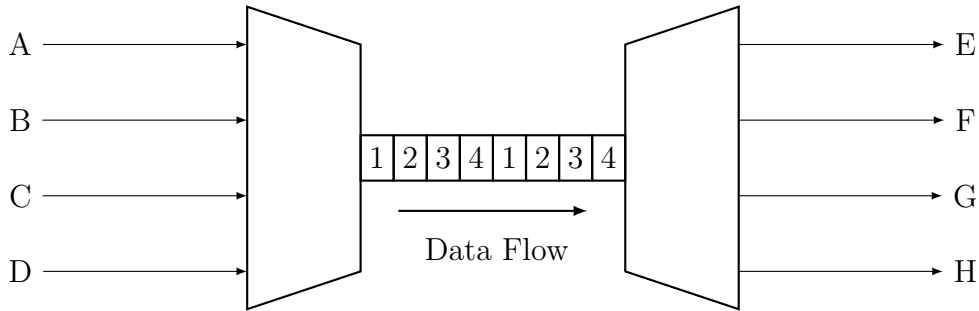
The illustration above shows a simple example of frequency-division into 5 channels, each with a capacity of 50 kHz and a 5 kHz gap between each channel. This ensures that when data is transmitted at a frequency lower or higher than the designated channel, it will not interfere with other channels.

4.2 Wavelength-Division Multiplexing

The Wavelength-Division Multiplexing mechanism is similar to Frequency-Division Multiplexing. Both FDM and WDM are very similar because they use composite signals; only the underlying physics is different. FDM is a general term used in radio-frequency communications, while WDM is used for fiber-optic media.

4.3 Time-Division Multiplexing

As we know from the name, time-division multiplexing is a multiplexing technique that determines which node sends data over a medium at a specific time. For example, Node A sends data during the time frame from a certain second to another, and the same applies to the other nodes. Consider the following illustration.



This illustration shows how dividing time into several frames can be implemented using TDM. Note that the small squares on the medium represent time intervals, and there are exactly 24 time slots. This means that within 1 unit of time, each node can transmit data twice.

For instance, if we define 1 unit of time as 1 second and each of the 4 nodes wants to transmit data exactly twice, then slot 1 is occupied by A, slot 2 by B, slot 3 by C, and slot 4 by D. Therefore, the time required to transmit 1 data packet in 1 window is

$$\frac{1 \text{ second}}{8 \text{ frame}} = 0.125 \text{ second}$$

Note: The Time-Division Multiplexing mechanism is divided into two types: Synchronous and Asynchronous TDM. What we just discussed is Synchronous TDM. Meanwhile, Asynchronous TDM is a mechanism in TDM which each transmitting node is not guarantee to get a time slot to send data.

4.4 Transmission Media

So far, we've talked a lot about how to share a medium when a number of nodes want to communicate. However, we haven't discussed the medium itself in much detail.

Let's discuss electromagnetic signal. According to 3rd and 4th Maxwell's equation:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (4)$$

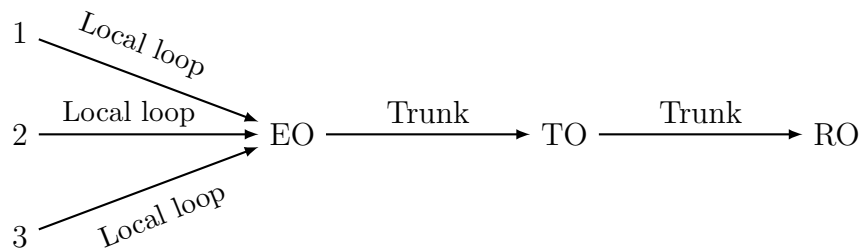
These two equations may seem complicated, but their meaning is simple. In (3), a curled electric field is proportional to the first derivative of magnetic field with respect to time (the electric field vector is orthogonal to the magnetic field). Meanwhile, in (4), a curled magnetic field can be caused by electric current density or the derivative of an electric field with respect over time by a factor of $\mu_0 \epsilon_0$, which is the constant resistance (the magnetic field vector is also orthogonal to the electric field). In summary: when there is an electric field, a magnetic field can be created, and vice versa.

Therefore, when we want to send data (a sequence of 0s and 1s) whether via a wired or wireless connection, the data transmitted from a smartphone or computer is converted into electromagnetic signals so that it can travel through cables or through the air. However, it's important to remember that electromagnetic waves can experience energy-loss, which means the information they carry can change at any time as it travels through the medium; that's why we to overcome to this problem by a technique called error detection (see Chapter 6).

Chapter 5

Network System

In this chapter, we will discuss the architecture of classic telephones, which were commonly used in the era of public payphones. Unlike the modern internet, which uses digital signals, classic telephones transmit voice information using analog signals.

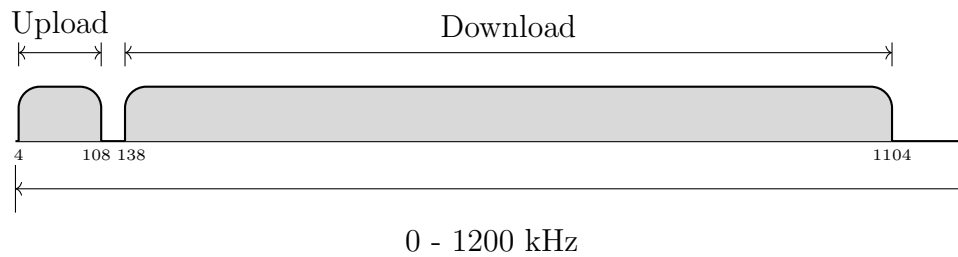


The illustration above shows that 1, 2 and 3 are the endpoints of each telephone connected to the End Office via the Local Loop. They are then connected via a trunk to the Tandem Office before connecting to the Regional Office.

The medium used in the local loop is twisted-pair cable with bandwidth a particular. Meanwhile, the trunk medium has greater capacity because it handles hundreds of phones through multiplexing.

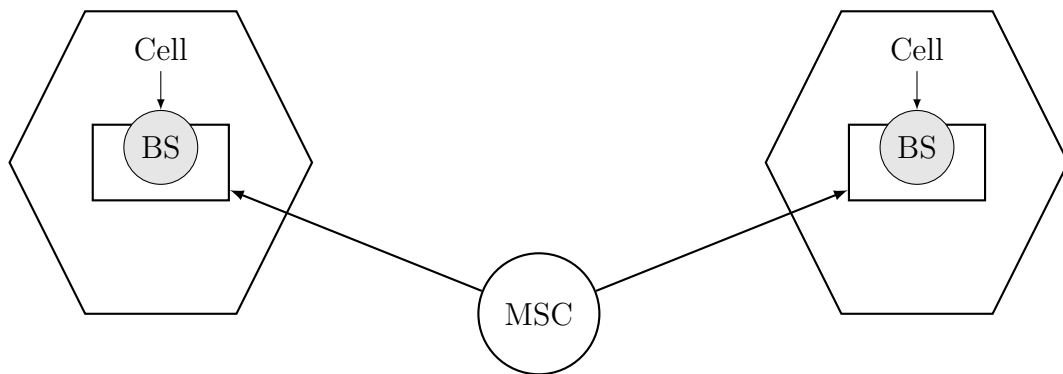
5.1 Digital-Transfer Line

Digital Transfer Line is a technology that enables high-speed signal transmission using copper cables, which were previously only used in telephone networks. One example of Digital Transfer Line is Asymmetric DSL. It is called “asymmetric” because across the entire frequency range, from the highest to the lowest frequencies, the download bandwidth is much greater than the upload bandwidth. This is due to the fact that users tend to download information from the internet more frequently than they upload it. Note the illustration below, which shows that the download bandwidth is significantly greater than the upload bandwidth in Asymmetric DSL.



5.2 Cellular Network System

This technology allows us to communicate via our smartphones wherever we go (moving nodes). Even though our location is constantly changing and we are not connecting our phones using cable over the time, we can still stay connected to the internet.



Consider the illustration above.

- i) Cell: A mobile phone that uses cellular data.
- ii) BS (Base Station): A data receiver from a cell, such as telecommunication tower. Keep in mind that the data transmitted from the cell to the BS is wireless.
- iii) MSC (Mobile Switching Center): A large hub that connects networks so that communication can take place even between different providers.

If you carry your smartphone (Cell) with you everywhere you go, the cell will search for the nearest cell tower (Base Station), and if it has trouble finding 1 or there isn't 1 at all (such as in the woods or in rural areas), you won't be able to access the internet.

Chapter 6

Data Link Layer

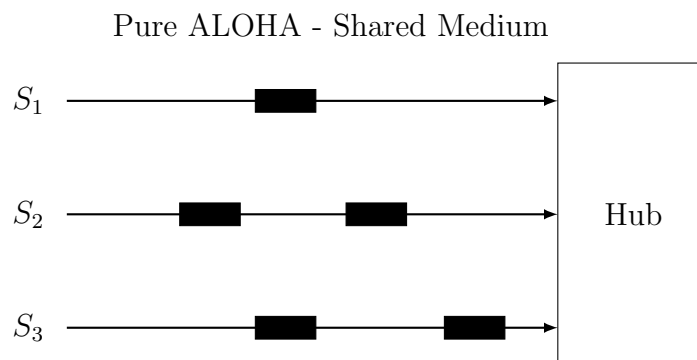
In this chapter, we will discuss the functions of the Data Link layer, which is the 6th layer of the OSI Model (see Chapter 2). To recall, the functions of the Data Link layer are:

- i) Access Control: Determines during which time slot we (as a node) are allowed to transmit data.
- ii) Flow Control: Determines how much data is sent during a single time slot.
- iii) Error Control: Handles errors if the data received by the receiver is incorrect.

6.1 Access Control

As we discussed earlier in the section on multiplexing (see Chapter 4), when we want to transmit data over a shared medium, there are two main methods; i.e. Frequency-Division Multiplexing (FDM) and Time-Division Multiplexing (TDM). Unlike FDM, which divides the medium into fixed channels, it is more difficult to divide time slots when using TDM. ALOHA and CSMA are two protocols commonly used in access control.

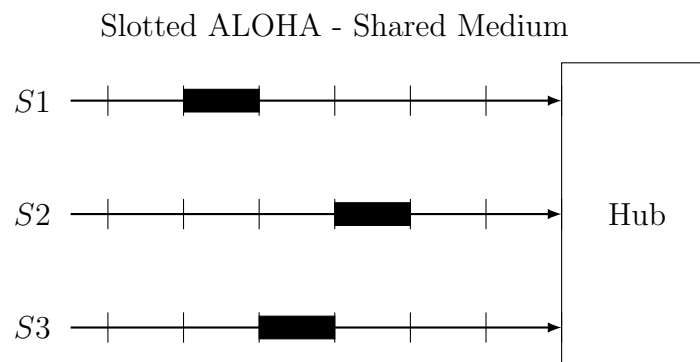
- i) ALOHA is divided into two types. It is Pure ALOHA and Slotted ALOHA. In Pure ALOHA, each node is allowed to transmit data at any time. Take a look at the following illustration of Pure ALOHA.



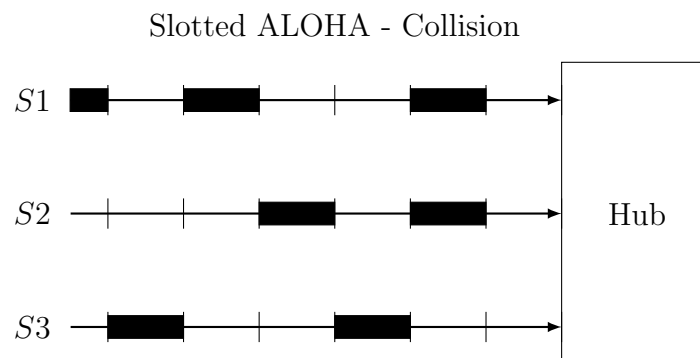
The black-horizontal rectangles represent packets sent by S_1 , S_2 , and S_3 . Note that on the same x -axis, there are two packets sent simultaneously by S_1 and S_3 . Neither of these packets will be received by the hub. Therefore, S_1 and S_3 must resend the packets that did not reach the hub.

How do S_1 and S_3 know that the packet they transmitted didn't actually arrive, and they have to resend it? The key is, for every packet successfully received by the hub, the hub sends an "Acknowledge" message back to the specific sender. Then happens is that, if the sender doesn't receive an acknowledgment, by default they have to resend the packet.

Slotted ALOHA is basically similar to Pure ALOHA. The key difference is that in Slotted ALOHA, time intervals for sending data are predetermined. Each sender may send data only at the beginning of an interval; they are not allowed to transmit data in the middle or at the end of the interval.



However, it is important to note that Slotted ALOHA does not guarantee that transmitted packets will be successfully sent without collision. Consider the following illustration.



As shown in the illustration above, there are slots occupied by two nodes. Therefore, acknowledgments are still required in Slotted ALOHA.

- ii) CSMA (Carrier Sense Multiple Access): A technique that requires the sender to check the medium's status (condition) before sending data. Basically, it tells the sender to not transmit if the medium is busy. Below is the formal algorithm for CSMA:

CSMA General Algorithm

```
1: while data is not transmitted yet do
2:   listen to medium
3:   if medium is idle then
4:     transmit data
5:     break
6:   else
7:     continue listening to medium
8:   end if
9: end while
```

CSMA/CD Algorithm

```
1: while data is not transmitted do
2:   if channel is busy then
3:     continue sensing channel
4:   else
5:     transmit data
6:     while transmission is in progress do
7:       if collision detected then
8:         send jamming signal
9:         stop transmitting
10:        wait for random backoff time
11:       break
12:     end if
13:   end while
14: end if
15: end while
```


CSMA/CA Algorithm

```

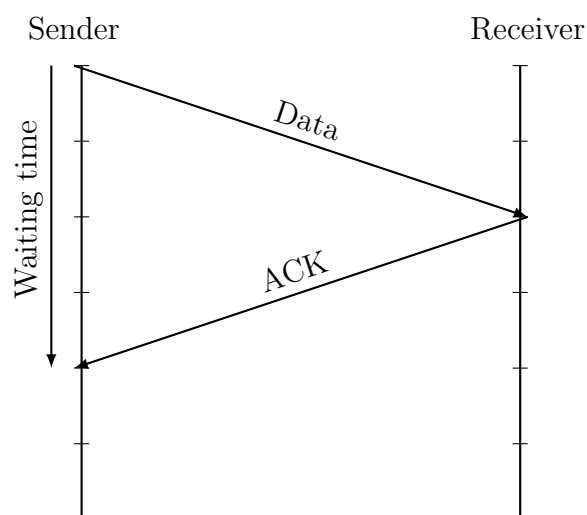
1: while data is not transmitted do
2:   if channel is busy then
3:     continue sensing channel
4:   else
5:     randomize backoff time
6:     wait for backoff time
7:     if channel is idle then
8:       transmit data
9:       wait for ACK
10:      if ACK received then
11:        break
12:      end if
13:    end if
14:  end if
15: end while

```

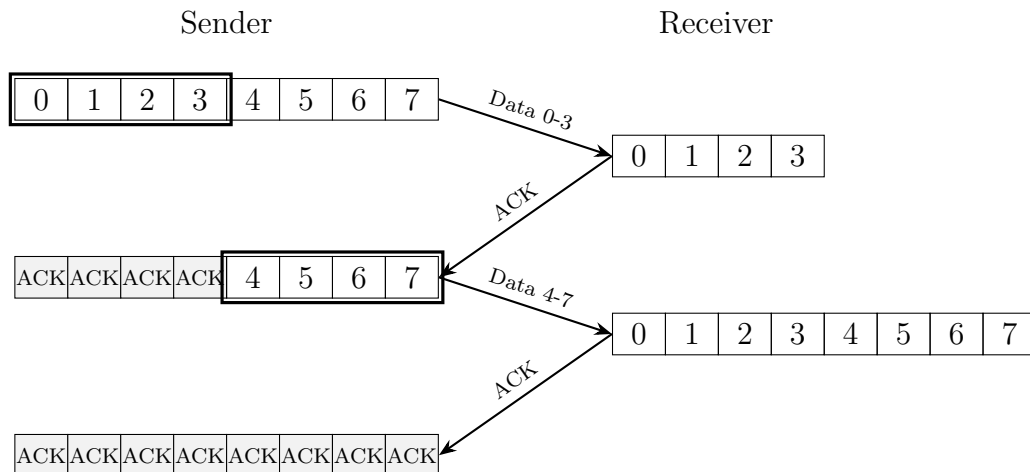
6.2 Flow Control

Flow control is a protocol procedure designed to control the amount of data transmitted in a single frame. There are two commonly used methods of flow control, namely:

- i) Stop-and-wait: This principle works by involving sending only 1 frame to a given window. The illustration below shows how the sender sends only 1 frame and waits to receive an acknowledgment.



- ii) Sliding window: Unlike the stop-and-wait method, the sliding window transmits multiple frames within a single, predetermined time slot. Consider the following illustration.

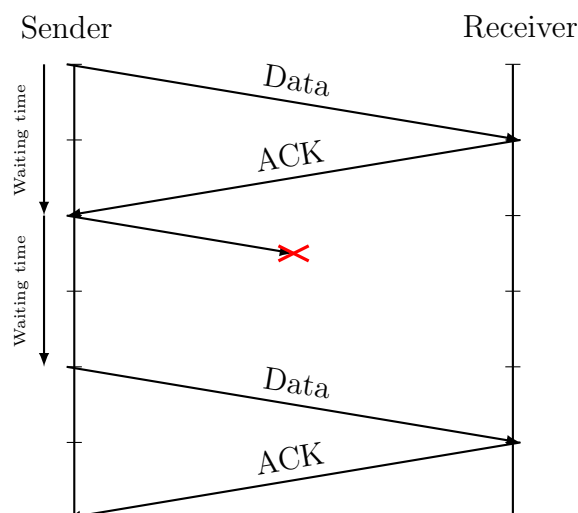


From the sliding window illustration, instead of sending 1 packet and receiving 1 acknowledgment at a time, the sender sends four packets and receives 4 acknowledgments all at once.

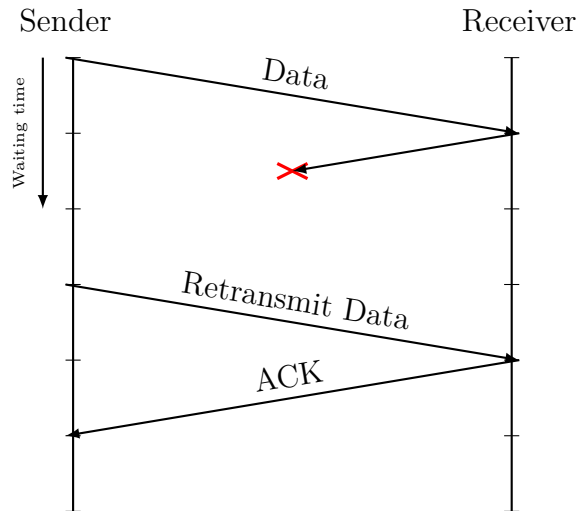
6.3 Error Control

The last fundamental function of the Data Link layer is to perform error correction. Error correction is a structured method for handling situations where a packet is not successfully received by the recipient (the sender does not receive an acknowledgment). One cause of this is energy loss in the medium. There are three techniques for handling situations where the sender does not receive an acknowledgment, namely:

- i) Stop and Wait: The simplest mechanism in error control, in which the sender waits for a specified period of time. If the sender waits too long without receiving an acknowledgment, it will attempt to resend the message.

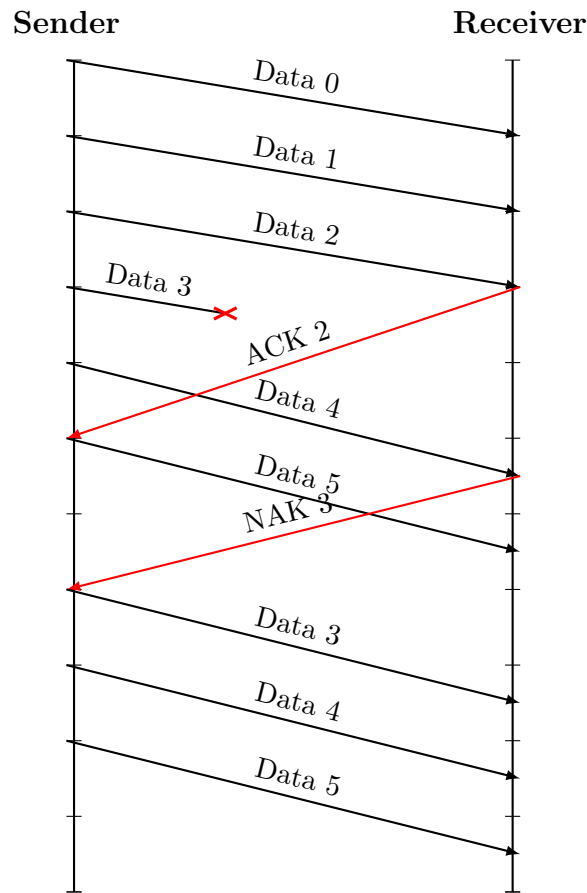


Note that the sender will wait for 2 time units (waiting time); if this time expires and the sender has not received an acknowledgment, the sender will resend the same data.



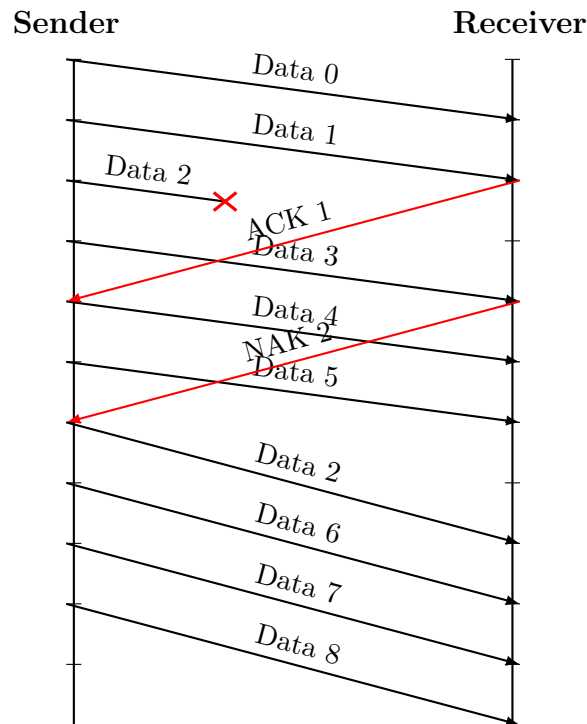
Instead of packet loss, an acknowledgment that never received by transmitter can also be caused by the acknowledgment itself being lost.

- ii) Go-Back-N: This error occurs only with the sliding-window packet mechanism in flow control.



The point of the illustration above is that when the sender transmits 4 packets within a single time window at once (still by order incrementally), and when the last data (number 3) in those packets is lost, the receiver will only acknowledge packets 0, 1 and 2. Even though the sender has not yet received acknowledgment 2, the sender will continue to send the next frame. Only when the receiver realizes that it received packet number 2 and then skipped to packet number 4 will it send a negative acknowledgment 3, which instructs the sender to retransmit from number 3 onward. Meanwhile, packet number 4, which the receiver has already received, will be discarded.

- iii) Selective Repeat: Unlike Go-Back-N error correction, Selective Repeat does not discard data that has already been transmitted and retransmit it. Instead, it only requests that the missing packets be retransmitted and continues transmitting from the last packet. Consider the following illustration.

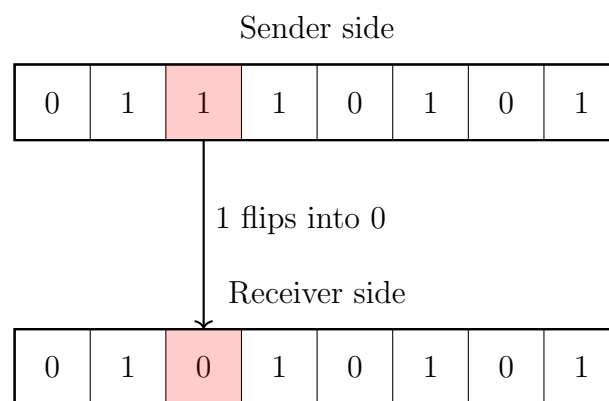


From the illustration above, after the sender successfully sending packets 0 and 1, and after the receiver sent an acknowledgment for frame 1, packet number 2 was lost. Since the flow control mechanism is a sliding window, the sender continues to send frame number 3 regardless of whether an acknowledgment has been received or not. However, the receiver realizes that after receiving packet number 1, it received packet number 3, meaning that packet number 2 never arrived. The receiver sends back a negative acknowledgment 2 to request packet number 2 again. But while the sender has not yet received the negative acknowledgment, it continues to send packets 4 and 5. Only after the sender receives negative acknowledgment 2 does it resend packet 2. Instead of having to resend packets 3, 4, and 5, the sender is allowed to continue sending from packet 6 onward because the receiver did not discard packets 3, 4, and 5.

Chapter 7

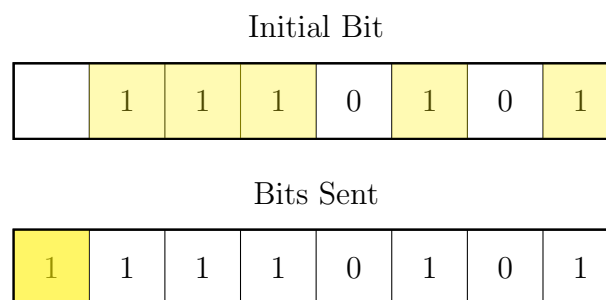
Error Detection and Correction

Following the discussion on error control in the previous chapter, which focused more on ‘what to do when data is lost’, in this chapter we will discuss how to detect data loss or data changes caused by the transmission medium. Keep in mind that a data error is not defined solely as data that does not actually reach the receiver; bits that are accidentally flipped are also considered as errors.



The illustration above shows a sender who wants to send the data ‘01110101’, but the receiver instead receives ‘01010101’. Of course, with data this small, it doesn’t seem like a big problem. However, if the sender sends larger amounts of data and errors occur repeatedly such as video, the received video will be unclear. There are four techniques for error detection and correction, namely:

- i) Parity Check: This technique adds a 1 to the most significant bit if the number of 1s in the data is odd, and adds a 0 to the most significant bit if the number of 1s in the data is even. Given the example:



The example above shows what happens when we want to send the bits ‘1110101’. The data consists of 7 bits, but we want to add a checksum for error detection. This means we add one bit at the beginning. That bit is set to 1 because there are five 1s in the ‘Initial Bit’. The bits that are actually sent are ‘11110101’.

The formal algorithm of Parity Check:

Parity Check Algorithm

```

1: for each byte blocks: do
2:   check how many 1s in this block
3:   if is in this block is even: then
4:     append 0 to the lefthand most bit
5:   else if 1s in this block is odd: then
6:     append 1 to the lefthandmost bit
7:   end if
8: end for
9: Send all modified bit block

```

- ii) Cyclic Redundancy Check: This technique uses the long division between two given bits. We will divide the bit we want to transmit by the ‘CRC polynomial’ bit. What we will transmit is the original data appended with the remainder of the division at the very end of the bit stream.

Consider

- a) Bits to be transmitted: 0110.
- b) CRC polynomial generated by the sender and receiver (both must be the same): $x^3 + x + 1$, or equivalently 1011 in bits.

Then we use the long division. Dividing bits to be transmitted by CRC Polynomial:

$$\begin{array}{r}
 \begin{array}{cccc} 0 & 1 & 1 & 1 \end{array} \\
 1011 \overline{) \begin{array}{ccccccc} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & & & \end{array}} & \leftarrow \begin{array}{l} \text{Add three 0s according} \\ \text{to the highest CRC degree} \end{array} \\
 \hline
 \begin{array}{cccc} 1 & 1 & 0 & 0 \end{array} \\
 \begin{array}{cccc} 1 & 0 & 1 & 1 \end{array} \\
 \hline
 \begin{array}{cccc} 1 & 1 & 1 & 0 \end{array} \\
 \begin{array}{cccc} 1 & 0 & 1 & 1 \end{array} \\
 \hline
 \begin{array}{cccc} 1 & 0 & 1 & 0 \end{array} \\
 \begin{array}{cccc} 1 & 0 & 1 & 1 \end{array} \\
 \hline
 \begin{array}{cccc} 0 & 0 & 1 & \end{array}
 \end{array}$$

The result of the long division is 0111. The data to be transmitted is:

- a) 0110001 (data to be sent + remainder)
- b) 0111 (result of the division).

Once the data has been received by the receiver, the receiver will verify it by dividing 0110001 by the CRC provided at the beginning (1011). Hence:

$$\begin{array}{rcccccccc}
& 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\
+ & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\
+ & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\
\hline
= & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1
\end{array} \tag{5}$$

When we take the one's complement of (5), we get

$$\text{Ones}(11111111) = 00000000 \tag{4}$$

If all of these algorithms have been run and the result is not 00000000, then a bit has changed on the medium (corrupted).

- iv) Hamming Code: Technique for performing corrections that are explicitly transmitted. If 7-bit data is provided:

1	0	1	1	0	0	0
7	6	5	4	3	2	1

We divide the 7-bit data into $\log_2 k + 1$ parts, where k is the length of the data in bits. If we have 7-bit data, then we divide the data into

$$\log_2 k + 1 = \log_2 8 = 3 \text{ parts}$$

Now we define P_1 , P_2 , and P_4 as the names of three groups. The n in P_n means that we take data from index n to the end, take n entries, and skip n entries. We do not want to include the very first data point, so we mark it with a question mark. Consider the following illustration:

	1	0	1	1	0	0	0
	7	6	5	4	3	2	1
$P_1 =$	1		1		0		?
$P_2 =$	1	0			0		?
$P_4 =$	1	0	1				?

From the illustration above, we can see that P_1 starts at index 1 and goes to the end, taking 1 entry and skipping 1 entry. Meanwhile, the very first entry is marked with a question mark. As for P_2 , we start at index 2, take 2 entries, and skip 2 entries (the data at index 2 is also marked with a question mark). Apply the same method to P_4 .

From all clusters in P_1 , P_2 , and P_4 , we obtain:

$$P_1 = 110, \quad P_2 = 100, \quad P_4 = 101$$

Next, we want to calculate the parity for each data group (see section i from this chapter for Parity Bits):

$$\textit{Parity}(P_1) = 0$$

$$\textit{Parity}(P_2) = 1$$

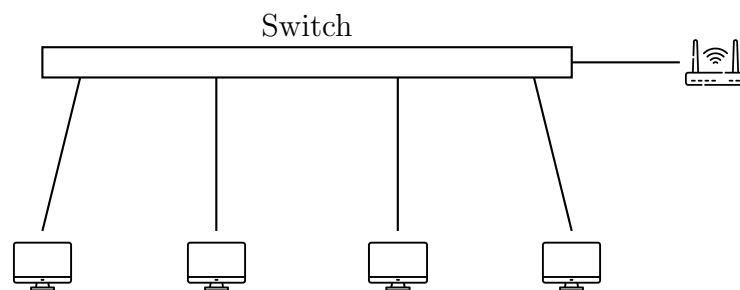
$$\textit{Parity}(P_3) = 0$$

If we combine the three bits, we obtain 010. Therefore, we know that an error occurred in the second bit because $010_2 = 2_{10}$.

Chapter 8

Local Area Network

As discussed in Chapter 1, a Local Area Network (LAN) is a network with a local scope. It typically covers only a few houses or a single housing complex. In other words, a Local Area Network is a system that allows nodes (computers) to communicate within a local area via wired or wireless medium. The following is a simple topology consisting of four computers connected via a switch and an access point, which allows smartphones to join the LAN via a wireless connection.



8.1 LAN Address

A LAN address, or MAC address, is an address assigned to every device (node) on a LAN and is used for communication within that local area. Its function is to serve as the device's identifier so that transmitted data doesn't get lost. A LAN address has the following two characteristics:

- i) The size of a LAN address $6 \text{ bytes} = 48 \text{ bits} = 12 \text{ hexadecimal}$
- ii) If the least-significant bit on the first byte
 - a) 0, then the device is a unicast device. This means that the device communicates with a single other device on the same network.
 - b) 1, then the device is a multicast device. This means that the device communicates with more than one other device on the same network.
- iii) If all hexadecimal values are FF, then the device is broadcasting. This means that the device is communicating with all other devices on the same network, without exception.

Example (in hexadecimal):

- i) The device with the address $4B : 20 : 8A : F1 : 0A : 1A$ is a unicast device, because the last two hexadecimal digits are $4A_8 = 01001010_{10}$. Since the last bit is 0, the device is a unicast device.
- ii) Perangkat yang memiliki address $FF : 2C : 1C : F3 : 4D : EE$ adalah unicast, karena 2 hexa terakhir adalah $FF_8 = 11111111_{10}$. Bit terakhir bernilai 1, maka perangkat itu adalah multicast.
- iii) A device with the address $FF : FF : FF : FF : FF : FF$ is a broadcast because all the hex digits are F.

8.2 Wireless LAN

A Wireless LAN is a Local Area Network that uses wireless devices such as smartphones and laptops. To set up a wireless LAN, you need an access point, as shown in the architectural diagram at the beginning of this chapter.

8.3 Bluetooth

Bluetooth is a specific technology that connects a number of devices within a smaller range than a LAN. Bluetooth is an “ad hoc” network, which means it does not require a fixed infrastructure to establish its architecture and can be set up spontaneously. With the latest technology, Bluetooth connectivity allows a device to connect to up to 8 other devices on a single network. There are two categories of Bluetooth networks, that is:

- i) Bluetooth Piconet: Consists of a master (controlling device) and a slave (controlled device). A piconet is further divided into two types, namely:
 - a) Point-to-point Piconet: One master that controls exactly one slave. One example is a smartphone connected to exactly one Bluetooth device, such as earbuds.
 - b) Point-to-multipoint Piconet: One master controls more than one slave. One example is a laptop connected to a Bluetooth webcam, Bluetooth speakers, a Bluetooth mouse, and a Bluetooth keyboard.
- ii) Bluetooth Scatternet: A Bluetooth connection in which a master has a slave, but that particular slave is also a master simultaneously. One example is when a laptop that has a Bluetooth mouse as a slave and a smartphone (for data sharing). In this case, the smartphone can be considered a slave to the laptop, but it is also a master if it has its own slaves, such as earbuds connected to it.

Chapter 9

Network Layer

This chapter discusses the functions of the Network Layer. Which is the fifth layer of the OSI model (see Chapter 2). As a reminder, the functions of the Network Layer are Internetworking, Addressing, Routing, Packetization, and Fragmentation. This chapter will focus on addressing and routing.

9.1 IP Datagram

An IP datagram is a collection of data consisting of an IP header + Main Data to be transmitted. The IP header contains important information about the main data, including the IP address, routing, addressing, and control information. The total size of an IP datagram is 65,535 bytes, with the IP header occupying only 20 bytes of the entire IP datagram.

9.2 IPv4 Address

An IPv4 address is $32 \text{ bits} = 8 \text{ hexadecimal} = 4 \text{ bytes}$. Since there are approximately 4 billion possible numbers in 32 bits , the number of devices that can be assigned an IPv4 address is roughly that many. However, data shows that the number of devices in the world exceeds the number of available IP addresses. To address this issue, some devices are designed to share their IPv4 addresses with other devices. Research into the latest IPv6 technology is still ongoing, and some systems are already using it.

Every IPv4 Address have the same structure, that is:

- i) Constant (Unique for every Class)
- ii) Network (Unique for every Organization)
- iii) Host (Unique for every User)

0	Network (7 bits)		Host (24 bits)		Class A		
1	0	Network (14 bits)		Host (16 bits)	Class B		
1	1	0	Network (21 bits)		Host (8 bits)	Class C	
1	1	1	0	Multicast		Class D	
1	1	1	1	0	Future Use		Class E

Take a look at the bar chart above. There are 5 classes of IPv4 networks, depending on who uses them. The purpose of class division is to group addresses based on network size. For instance, Class A can accommodate more devices because its identifier constant is only 1 bit. Class A is usually used by international networks or multinational telecommunication companies. Different networks within the same class (such as in Class A) represent the number of organizations using that class. The total number of potential users who can use this class is $2^{24} \approx 16 \text{ million hosts}$. Meanwhile, the number of networks that can use Class A is $2^7 = 128 \text{ networks}$. This means there are 128 different organizations that can use this class. For each network or organization, there are approximately 16 million users. This means the total number of users who can use this Class is $2^{24} \cdot 2^7 \approx 2.1 \text{ billion users}$. The same explanation applies to Classes B, C, D, and E.

9.3 Subnet Mask

We now know the function and mechanism of class division in IPv4. So how do routers or other devices know which bits are network bits and which are host bits? The answer is through subnetting. The format of a subnet mask is the same as that of an IP address: $32 \text{ bits} = 8 \text{ hexadecimal} = 4 \text{ bytes}$. Here is an example of a simple subnet mask with 3 different notations:

255.255.255.0 (Decimal notation)

(IPAddress)/24 (CIDR notation)

11111111.11111111.11111111.0 (Binary notation)

The easiest way to visualize a subnet mask is in binary notation. If the entry is 1, then the corresponding bit position in the IP address represents the network.

Meanwhile, the corresponding bit position is the host when the subnet mask is 0. For example, if we are given a complete address with a subnet mask in CIDR notation in the format [IP Address]/[Subnet Mask]:

192.168.1.254/26

We convert from decimal notation to binary notation give more intuitive visualization.

11000000.10101000.00000001.11111110 (IP Address)

As for the subnet mask, we know it in CIDR notation. To convert it to binary notation, we set the first 26 digits to 1, while the rest are 0. This gives us the CIDR /26 as:

11111111.11111111.11111111.11000000 (Subnet Mask)

The 0 entry in the subnet mask represents the host bits (i.e., the last 6 digits), while the first 26 digits represent the host with the Class constant discussed earlier.

9.4 Routing Algorithm

A routing algorithm is an algorithm that finds the shortest path for data sent from a sender to a receiver. There are two common algorithms used for routing: Distance Vector Routing (Bellman-Ford's Algorithm) and Link-State Routing (Dijkstra's Algorithm). The formal algorithm for Bellman-Ford's Algorithm is:

Bellman-Ford's Algorithm

```

1: for all node  $x$  do
2:   set distance to itself as 0
3:   set distance to adjacent nodes according to link cost
4:   set distance to non-adjacent nodes as  $\infty$ 
5:   send distance vector to all adjacent nodes
6: end for
7: while distance table changes do
8:   for all node  $x$  do
9:     for all destination node  $y$  do
10:       $D_x(y) \leftarrow \min_{u \in Neighbor(x)} [c(x, u) + D_u(y)]$ 
11:    end for
12:    send updated distance vector to all adjacent nodes
13:  end for
14: end while

```

Chapter 10

Multicast Address Class

To recall, multicast is a telecommunication method in which a single node sends information to more than one node. In a single multicast device, there are two addresses, namely:

- i) IP Address: Has a 32-bit capacity.
- ii) MAC (Multicast Address Class): Has a 48-bit capacity. It consists of a 25-bit constant at the beginning (01:00:5E in hexadecimal). The final 23 bits have the same values as the final 23 bits in the IP address.

For instance, if we want to find the MAC address when the IP address of a multicast device is known:

$$11001011 : 00101100 : 10011000 : 00111001 \quad (1)$$

Hence we take the last 23-bit from (1):

$$0101100 : 10011000 : 00111001 \quad (2)$$

We consider the MAC constant with the value 01:00:5E and convert it to binary:

$$01 : 00 : 5E_8 = 00000001 : 00000000 : 01011110 : 0_2 \quad (3)$$

If we combine the 23-bits from (2) and the 25-bits from (3), we will get a 48-bit MAC address:

$$00000001 : 00000000 : 01011110 : 00101100 : 10011000 : 00111001 \quad (4)$$

Formally, we can convert (4) to a hexadecimal address

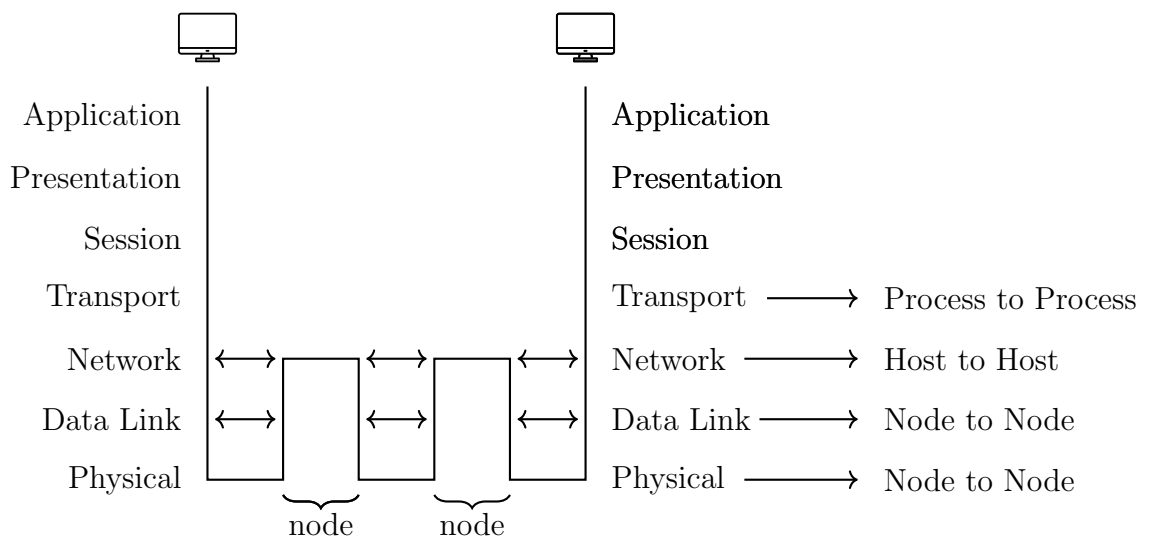
$$01 : 00 : 5E : 2C : 98 : 39 \quad (5)$$

This means that if a multicast device has the IP address $CB : 2C : 98 : 39$, its MAC address is $01 : 00 : 5E : 2C : 98 : 39$.

Chapter 11

Transport Layer

This chapter will discuss the functions of the Transport Layer, which is Layer 4 of the OSI Model.



From the illustration above, we can see that the Data Link and Network layers operate at every node. This means that tasks from Routing to Flow Control is performed at each node (such as a router that forwards data from the sender to the receiver). Meanwhile, the Transport Layer operates only at the sender and receiver themselves, which means the Transport Layer operates end-to-end.

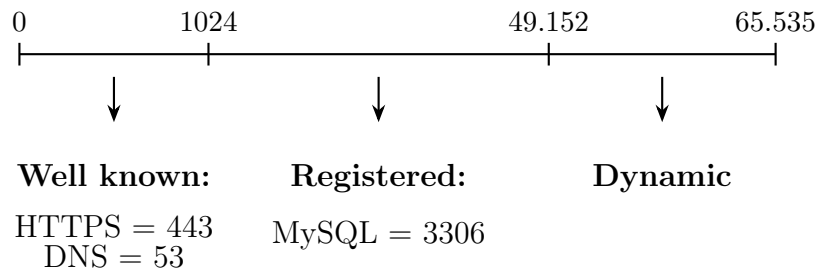
11.1 Process-to-process

Once the data has been successfully received by the receiver, the computer needs to know which program on the receiver, the data belongs to. In other words, we need to create specifications or additional data that indicate which program the received data should be forwarded to. This process uses a principle similar to multiplexing, where the data received by the receiver is distributed across the 'medium' even while it is still on the receiving device itself.

11.2 Port Number

Let's define a Port Number. It is a data header that informs the receiver which program this data belongs to. Without a port number, our computer wouldn't know which program the data received from another computer is intended for. For instance, suppose we receive incoming email data from another device. How would our computer know that this data belongs to an incoming email? The computer might interpret the data as being for YouTube or TikTok instead. That's where the port number comes in to provide information about the program associated with that data. If the IP address is information about "Which computer should we send this data to?", then the port number is "To which program should we forward this data?".

IANA (Internet Assigned Numbers Authority) is the standard for assigning port numbers in every program. Here is an illustration with some examples of common port uses (HTTPS, DNS, MySQL):



Note: A socket address is a unique address assigned to each piece of transmitted data to be sent, consisting of an IP address and a port number. Its purpose is to provide a complete destination for the data being sent, down to the program level (process-to-process).

Two Common Protocols in Process-to-Process Communication:

- i) UDP (User Datagram Protocol): A packet that will later be encapsulated within an IP packet. The contents of the UDP header consist of:
 - a) Source Port Number (2 bytes): Sender's port
 - b) Destination Port Number (2 bytes): Receiver's port
 - c) Total Length (2 bytes): The size of the data being sent
 - d) Checksum (2 bytes): Error correction (see Chapter 7)

These four components will form the header of the transmitted data to be sent and forwarded to the next layer (the Network layer, if transmitting) for further data transmission or reception.

- ii) TCP (Transmission Control Protocol): A connection-oriented protocol. This means that both parties establish a connection to ensure end-to-end communication.

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